

HAYDEN FU

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PROFESSIONAL SUMMARY

Computer Science and Statistics student at Cornell University with experience building data pipelines and applying statistical and machine learning methods to heterogeneous, real-world datasets. Skilled in Python, probabilistic modeling, and forecasting.

EDUCATION

Cornell University, College of Arts & Sciences Expected May 2028
B.S., Computer Science and Statistics
Relevant Coursework: Probability Models & Inference, Applied Linear Statistics, Categorical Data Analysis, Machine Learning, Algorithms, Computer Architecture, Functional Programming, Discrete Math, Linear Algebra, Multivariable Calculus

EXPERIENCE

Algoverse LLM Research Group | *Applied AI Engineer* May 2026 – Present

- Built a time-aligned LLM forecasting pipeline (Python, Azure AI Foundry) combining prediction-market prices, order-book data, and news evidence to measure model calibration and drift.
- Evaluated 3 open-weight LLMs across 100 sequential forecasting tasks using linear probes and counterfactual interventions; resulting paper accepted to the IAB workshop at NeurIPS 2026.

Cornell University, Civil Engineering | *Data Specialist* Jun 2026 – Present

- Built a particle-filter pipeline in Python (xarray, pandas, NetCDF, scikit-learn) for probabilistic inference over large-scale, multi-source time-series data.
- Designed preprocessing and calibration workflows to align noisy satellite observations, model outputs, and geospatial boundaries across multi-day inference windows.
- Leveraged Claude Code to debug and refactor the pipeline's diagnostics, improving readability and reproducibility.

Cornell Cybersecurity Club | *Competition Team Member* Sep 2025 – Present

- Competed in international Capture-the-Flag competitions; placed 1st at BSides NYC CTF (also AmateursCTF, OSU CTF, Buckeye CTF).

Cornell Geo Data | *ML Research Engineer* Nov 2025 – Present

- Built a GAN-based corrective ML model (Python, PyTorch) to improve short-term forecast accuracy using sensor and numerical prediction data.
- Cleaned and validated raw sensor time-series data to support model evaluation.

Harvard Institute of Quantitative Social Sciences | *Research Intern* Jun 2024 – Aug 2024

- Applied difference-in-differences regression to housing and policy data to evaluate causal effects; published in the Journal of Student Research.

University of Illinois Urbana-Champaign, College of Education | *Applied AI Developer* Sep 2023 – May 2024

- Built a data-driven map app (Python, ChatGPT-powered classification pipeline) giving K-12 educators access to 1,000+ resources.

PROJECTS

Six Hops | BigRed//Hacks 2026 | **Capital One & Solana Track Winner** Oct 2026

- Co-developed referral-network visualizations and analyzed 1M simulated events with SQL dashboards to evaluate search propagation and reward strategies.
- Built a full-stack referral search platform (Next.js, TypeScript, PostgreSQL) integrating iMessage workflows and automated Solana devnet payouts.

SKILLS

Languages: Python, SQL, R, TypeScript, Java, JavaScript, C/C++, OCaml, LaTeX
Data & ML: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, matplotlib, xarray, NetCDF
Methods: Regression analysis, hypothesis testing, probabilistic inference, statistical modeling, data pipeline design
Tools: PostgreSQL, Git, Claude Code, Azure AI Foundry, Jupyter, Linux/Unix